

External Validation One-Page

MIDAS Section Library

embedded metadata validated

static + contract gate

MIDAS section-library: ok | 182/183 used | 183 templates | source=midas_parser_derived |
implementation/phase1/open_data/midas/midas_generator_33.json
nightly, release gap, committee dashboard, and this external validation onepage all consume the same validator line

Validator line

MIDAS section-library: ok | 182/183 used | 183 templates | source=midas_parser_derived |
implementation/phase1/open_data/midas/midas_generator_33.json

Bundle	20260422T140547Z
Generated	2026-04-22T14:05:48.342988+00:00
Nightly	PASS
CI	PASS
Static validation	PASS
Freeze	PASS
Promotion	PASS
Promotion reason	PASS
Promotion hold	False
Hold review manifest	implementation/phase1/release/hold_review_manifest.json
Hold review packet	implementation/phase1/release/hold_review_packet.md

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Hold review packet pdf	implementation/phase1/release/hold_review_packet.pdf
Hold review ack	implementation/phase1/release/hold_review_ack.json
Smoke reason	PASS
Smoke pass rate	100.00%
Smoke trial feasible	100.00%
Smoke avg trial runtime (s)	0.0563
Smoke strict recommendation	candidate_for_strict_enable
Signed registry	PASS
Registry signature	PASS
Solver HIP e2e	PASS
RC benchmark lock	PASS
NDTHA residual gate	PASS
Committee package	PASS
MIDAS section-library validator	MIDAS section-library: ok 182/183 used 183 templates source=midas_parser_derived imple
MIDAS section-library status	embedded metadata validated
Material constitutive	Material constitutive gate: PASS concrete_damage=yes(matrix=48/48,max=1.000) cyclic_deg
Surface interaction benchmark	Surface interaction benchmark: PASS ready=7/7 family_matrix=420/420 source_families=10
KDS row provenance export	MIDAS KDS row provenance export: PASS combos=6 rows=144 members=12 clauses=6 e

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KR ingest	KR ingest: PASS src=15 cls=4 got=10 fp=10 meta=5 rej=0 dup=0 seed=15 topo=5 n
Measured benchmark breadth	Measured benchmark breadth: PASS baseline=2/51 opensees_delta=6/7 authority_delta=2/6
OpenSees canonical breadth	OpenSees canonical breadth: PASS families=6 cases=7 parser_ready=3 origins=designsafe_
KR preview queue	KR preview queue: PASS cand=8 pend=0 state=closed_until_new_public_archive_exact_topo
KR representative batches	building=kci_concrete_building_design_examples_2e_native_baseline housing=lh_bucheon_ye
Irregular structure track	Irregular structure collection gate: PASS families=20 sources=32 local_ready=15 remote_ca
Irregular benchmark execution	Irregular benchmark execution: CHECK ready=0 blocked=0 task_count=0 canonical=0 bric
Commercial grade	Commercial
Deployment model	engineer_in_the_loop_accelerated_coverage
Accelerated coverage	95-99%
Residual holdout	1-5%
Estimated time saved	90-96%
Empirical smoke runtime reduction	95.66-96.04%
Comparable run mode	current_pipeline_comparable_full_chain_pass
Comparable deployment	engineer_in_the_loop_accelerated_coverage
Comparable strict smoke	True
Engineer-in-loop ready	True
Estimated time basis	Empirical estimate derived from nightly design-optimization smoke runtime reduction, scaled by

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Time-saving focus	Use this engine to automate the dominant, time-consuming 95-99% of repeated analysis, screening
Full replacement ready	False
External benchmark start now	True
External benchmark full submission ready	True
External benchmark reason	PASS_START_NOW_FULL
External benchmark start mode	start_now_full_external_submission
External benchmark scope	full_external_submission_package
External benchmark blockers	none
External benchmark cautions	panel_zone_external_validation_only_boundary
External benchmark execution mode	limited
External benchmark execution ready tasks	10
External benchmark execution blocked tasks	0
External benchmark execution review-boundary	0
External benchmark review-boundary resolution	approve_all=PASS_NO_OPEN_DECISION_ITEMS/ready_full=yes; reject_one=PASS_NO_OPEN_D
External benchmark execution status	execution_complete_no_fail
External benchmark execution status counts	planned=0 in_progress=0 completed=10 failed=0 finished=10 completion_ratio=1.000
External benchmark case attestation workflow	cases=10 manifests=10 templates=0 receipts=10 attested=10 source=manifest=10 statu
PBD response source	resolved_report=implementation/phase1/experiments/by_test/nonlinear_ndtha_stress/2026031

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Audit review decision batch template	items=0 status=none owner=none priority=none attested_examples=0 example_preview=
Approve-all readiness preview	reason=PASS_NO_OPEN_DECISION_ITEMS ready_full=False pending=0 open_revision=0
Reject-one readiness preview	reason=PASS_NO_OPEN_DECISION_ITEMS ready_full=False pending=0 open_revision=0 bl
Audit review decision batch runner	reason=PASS_ZERO_TOUCH_NO_OPEN_DECISION_ITEMS apply_live=False live_applied=False
Structural optimization viewer	implementation/phase1/release/visualization/structural_optimization_viewer.html mode=static
Optimized drawing review	implementation/phase1/release/visualization/optimized_drawing_review.html projections=3 c
Authority routing warning	none
Open gaps	P0=0 / P1=0 / P2=0
MIDAS rows	total=12728, skipped=0, unknown=0
KDS rows	compliance=511, member_checks=1056, clauses=16
NDTHA residual	top=0.6880580090188355 m, drift=1.9136000000000006 %, fallback_rate=0.0
Registry artifacts	12
Design-opt long feasible	True
Design-opt long final max DCR	0.9331636363636363
Design-opt raw max drift	9.200000000000003
Design-opt repaired compliance max drift	1.9872000000000005
Design-opt compliance basis	repaired_solver_validated_slice
Design-opt repair action count	1

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Design-opt constructability signal gain	0.6812506971740343
Design-opt constructability avg	0.32310580388643245 -> 0.32090464334484636
Design-opt detailing complexity avg	0.4242652309769553 -> 0.4227255791394311
Design-opt selected family mix	connection_detailing=5, detailing=6, perimeter_frame=1, rebar=2, slab_thickness=2, wall_thick
Design-opt selected dominant family	wall_thickness (30.43%)
Design-opt selected family mix trend	connection_detailing=+0, detailing=+0, perimeter_frame=+0, rebar=+0, slab_thickness=+0, wa
Design-opt previous dominant family	wall_thickness (30.43%)
Design-opt preview supply family mix	beam_section=7, connection_detailing=14, detailing=142, perimeter_frame=1, rebar=254, slab
Design-opt preview missing target families	
MGT export direct-patch changes	25
MGT export direct-patch families	beam_section=1, connection_detailing=6, detailing=5, perimeter_frame=1, rebar=5, slab_thick
MGT rebar namespace mode	group_local
MGT rebar delivery mode	direct_patch_eligible
MGT evidence model	direct_patch_plus_zero_touch_verification_manifest
MGT rebar material namespace present	True
MGT rebar group-local namespace present	True
MGT material rebar payloads	3/5
MGT group-local rebar payloads	6

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MGT rebar direct-patch eligible	6
MGT rebar direct-patch blockers	
MGT rebar mapping sources	alt_slab_wall_group_id=5, direct_group_id=1
Design-opt cost delta	411.15925674073515
Design-opt changed groups	20
Blocked cost-down rows	47
Blocked illegal-by-mask	0
Illegal-by-mask families	
Blocked no-cost-gain	8
Blocked constructability hard-gate	3
Hard-gate reasons	detailing_ratio_above_hard_limit=3
Hard-gate families	beam_section=1, rebar=2
No-cost-gain groups	5
No-cost-gain explain rows	5
Design-opt groups	Ablation=1/1 Dataset=1/1 Profile=1/1 Stage A=2/2 Stage A/B/C=1/1 Stage B=1/1
Frame metrics	cases=3, drift_p95=4.167%, top_disp_p95=3.485%
Wind metrics	cases=4, max_drift=0.000806%, residual_drift=0.000113%
SSI metrics	cases=4, nonlinear_span=0.282342, residual_drift=0.004034%

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Design-opt metrics

rows=13, raw_drift=9.2000%, repaired_drift=1.9872%, cost_delta=411.159

This is the current external-validation submission baseline. Previous submission bundles were pruned after this package was created.

Appendix: MIDAS Native Roundtrip / Write-Back Taxonomy

```
- roundtrip_gate_summary: `MIDAS native roundtrip: PASS | corpus=77 | native_text=35 | public_native=9 | public_raw_native=8 | public_bridge_native=1 |  
public_preview_native=7 | public_structural_preview_native=8 | fixture_native=4 | repo_native=3 | experiment_native=3 | archives=7 | ready=35 | public_ready=25 |  
public_native_ready=9 | public_raw_ready=8 | public_bridge_ready=1 | public_archive_ready=7 | public_structural_preview_ready=8 | exact_queue=0/8 |  
whiteback_diff_summary: `MIDAS-native write-back diff receipts: PASS | ready=35/35 | receipts=35/35 | topology=35/35 | load=35/35 | loadcomb=35/35 | exact|types=12 |  
korean_reconstruction=35 | korean_promotions=0/0 | texture_ready=4 | repo_ready=3 | experiment_ready=3 | receipts=35/35 | topology=35/35 | load=35/35 | loadcomb=35/35  
exact|coverage-ready:100%|receipts:100%|loadcomb:100%|public_source:71%|exact_queue:100%|kr_recon:100% |  
gap_synchrize_receipts:0/0|texture_ready:4|repo_ready=3|experiment_ready=3|public_native=9 | public_raw_native=8 | public_bridge_native=1 | public_preview_native=7 | public_structural_preview_native=8 |  
taxidom_exact=25/34|structural_promosy=12 | batches=12 | receipts=35/35 | pending_review=0  
taxonomy_exact=34|canonicalPromosy=0 | unsupported_manual=0 | pending_review=0
```

```
- `public_split`: public_native_ready=9 | public_raw_ready=8 | public_bridge_ready=1 | public_archive_ready=7 | public_structural_preview_ready=8 |  
public_source_ready=25 | fixture_ready=4 | repo_ready=3 | experiment_ready=3
```

```
- `kr_preview_promoted`: ready=8 | cand=8 | addl_now=0
```

```
- 'special_member_family': direct_patch='core_beam_connection_detailing=1, intermediate_beam_section=1, intermediate_wall_rebar=4, intermediate_wall_thickness=3,
perimeter_frame_column=1, perimeter_slab_thickness=2, perimeter_wall_detailing=5, perimeter_wall_rebar=1, perimeter_wall_thickness=2,
transfer_beam_connection_detailing=5' | supported='core_beam_connection_detailing=2, intermediate_beam_section=1, intermediate_wall_rebar=4,
intermediate_wall_thickness=3, perimeter_frame_column=1, perimeter_slab_thickness=2, perimeter_wall_detailing=10, perimeter_wall_rebar=1, perimeter_wall_thickness=2,
markdown='implementation/phase1/release/midas_native_roundtrip/exact_topology_structural_preview_promotion_queue.json' |
transfer_beam_connection_detailing=10' | zero_touch_verified='core_beam_connection_detailing=1, perimeter_wall_detailing=5, transfer_beam_connection_detailing=5'
- 'artifacts': gate_json='implementation/phase17/midas_native_roundtrip_gate_report.json' |
receipts_report='implementation/phase1/release/midas_native_roundtrip/midas_native_writeback_diff_receipts_report.json' |
appendix_md='implementation/phase1/release/midas_native_roundtrip/unsupported_lossy_card_family_appendix.md' |
appendix_json='implementation/phase1/release/midas_native_roundtrip/unsupported_lossy_card_family_appendix.json'
- 'promotion_queue_state': 'No supported exact topology archive candidates are pending right now. The queue reopens automatically when a new public archive decoded
preview lands with exact_topology_candidate=true.'
```

```
- taxonomy case counts: {"canonical_rewrite": 1, "lossy_rewrite": 0, "manual_review_required": 0, "parser_drop_suspected": 1, "preserved_exact": 34, "unsupported_card": 0}
```

```
- taxonomy_card_family_histogram: {"audit_manifest_action_families": {}, "audit_only_action_families": {}, "direct_patch_action_families": {"beam_section": 1, "connection_detailing": 6, "detailing": 5, "perimeter_frame": 1, "rebar": 5, "slab_thickness": 2, "wall_thickness": 5}, "supported_action_families": {"beam_section": 1, "connection_detailing": 12, "detailing": 10, "perimeter_frame": 1, "rebar": 5, "slab_thickness": 2, "wall_thickness": 5}, "unsupported_reason_counts": {}}
```

| Structure Type | Ready | Receipts | Topology | Load | LoadComb | Pending Review | Batch Markdown |

$$| \text{---} | \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} |$$

```
| beam | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas_native_roundtrip/beam.diff_batch.md |
```

```
| bearing | 1 | 1 | 1 | 1 | 1 | 0 | implementation/phase1/release/midas_native_roundtrip/bearing.diff batch.md |
```

```
| bridge | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas_native_roundtrip/bridge.diff_batch.md |
```

```
| bridge_section | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas_native_roundtrip/bridge_section.diff_batch.md |
```

```
| building | 11 | 11 | 11 | 11 | 11 | 0 | implementation/phase1/release/midas_native_roundtrip/building.diff_batch.md |
```

```
| foundation | 4 | 4 | 4 | 4 | 4 | 0 | implementation/phase1/release/midas_native_roundtrip/foundation.diff_batch.md |
```

```
| housing | 1 | 1 | 1 | 1 | 1 | 0 | implementation/phase1/release/midas_native_roundtrip/housing.diff_batch.md |
```

[illegible]

[illegible]

Appendix: Irregular Structure Track

- `track\_summary`: `Irregular structure collection gate: PASS | families=20 | sources=32 | local\_ready=15 | remote\_candidates=17 | collected=21 | top5=5`
- `benchmark\_manifest`: `implementation/phase1/release/external\_benchmark\_kickoff/irregular\_benchmark\_execution\_manifest.json`
- `counts`: `families=20 | sources=32 | local\_ready=15 | remote\_candidates=17 | collected=21 | native\_roundtrip\_candidates=18 | solver\_candidates=13 | ai\_candidates=32 | top5=0`
- `top5\_split`: `local\_ready=0 | remote\_needed=0 | family\_ids=n/a`
- `benchmark\_execution\_summary`: `Irregular benchmark execution: CHECK | ready=0 | blocked=0 | task\_count=0 | canonical=0 | bridged=0 | proxy=0 | receipts=0`

Appendix: MIDAS KDS Row Provenance Export

MIDAS KDS row provenance export: PASS | combos=6 | rows=144 | members=12 | clauses=6 | exact_rows=144 | clause_filters=6 | member_filters=12 | reverse_jump=viewer_subset_reverse_jump_v11

json=implementation/phase1/release/kds_compliance/midas_kds_row_provenance_table.json |
csv=implementation/phase1/release/kds_compliance/midas_kds_row_provenance_table.csv |
report=implementation/phase1/release/kds_compliance/midas_kds_row_provenance_table_report.json

gLCB1 | member=MF-012 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

gLCB1 | member=MF-005 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member_types=column

gLCB1 | member=MF-012 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

gLCB1 | member=MF-010 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-001 | case=C-TST-001 | baseline=27441 | member_types=column

gLCB1 | member=MF-012 | clause=KDS-SHEAR-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

gLCB1 | member=MF-005 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member_types=column

Appendix: MIDAS KDS Row Provenance Export

gLCB1 | member=MF-007 | clause=KDS-MOMENT-Y-001 | baseline=27425

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-007 | case=C-TRN-007 | baseline=27425 | member_types=wall

gLCB1 | member=MF-012 | clause=KDS-AXIAL-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member_types=column

clause=KDS-AXIAL-001 | rows=24 | members=12 | combos=2 | top_member=MF-012 | top_dcr=1.065

clause=KDS-INT-FRAME-001 | rows=24 | members=12 | combos=2 | top_member=MF-012 | top_dcr=1.137

clause=KDS-MOMENT-Y-001 | rows=24 | members=12 | combos=2 | top_member=MF-012 | top_dcr=1.216

clause=KDS-MOMENT-Z-001 | rows=24 | members=12 | combos=2 | top_member=MF-012 | top_dcr=0.991

clause=KDS-SHEAR-Y-001 | rows=24 | members=12 | combos=2 | top_member=MF-012 | top_dcr=1.110

clause=KDS-SHEAR-Z-001 | rows=24 | members=12 | combos=2 | top_member=MF-012 | top_dcr=0.863

member=MF-001 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

member=MF-002 | baseline=27287 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

member=MF-003 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

member=MF-004 | baseline=27425 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

member=MF-005 | baseline=27441 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

member=MF-006 | baseline=27287 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

Appendix: MIDAS KDS Row Provenance Export

member=MF-007 | baseline=27425 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

member=MF-008 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001

hazard=combined | rows=48 | members=4 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001 | top_dcr=1.216

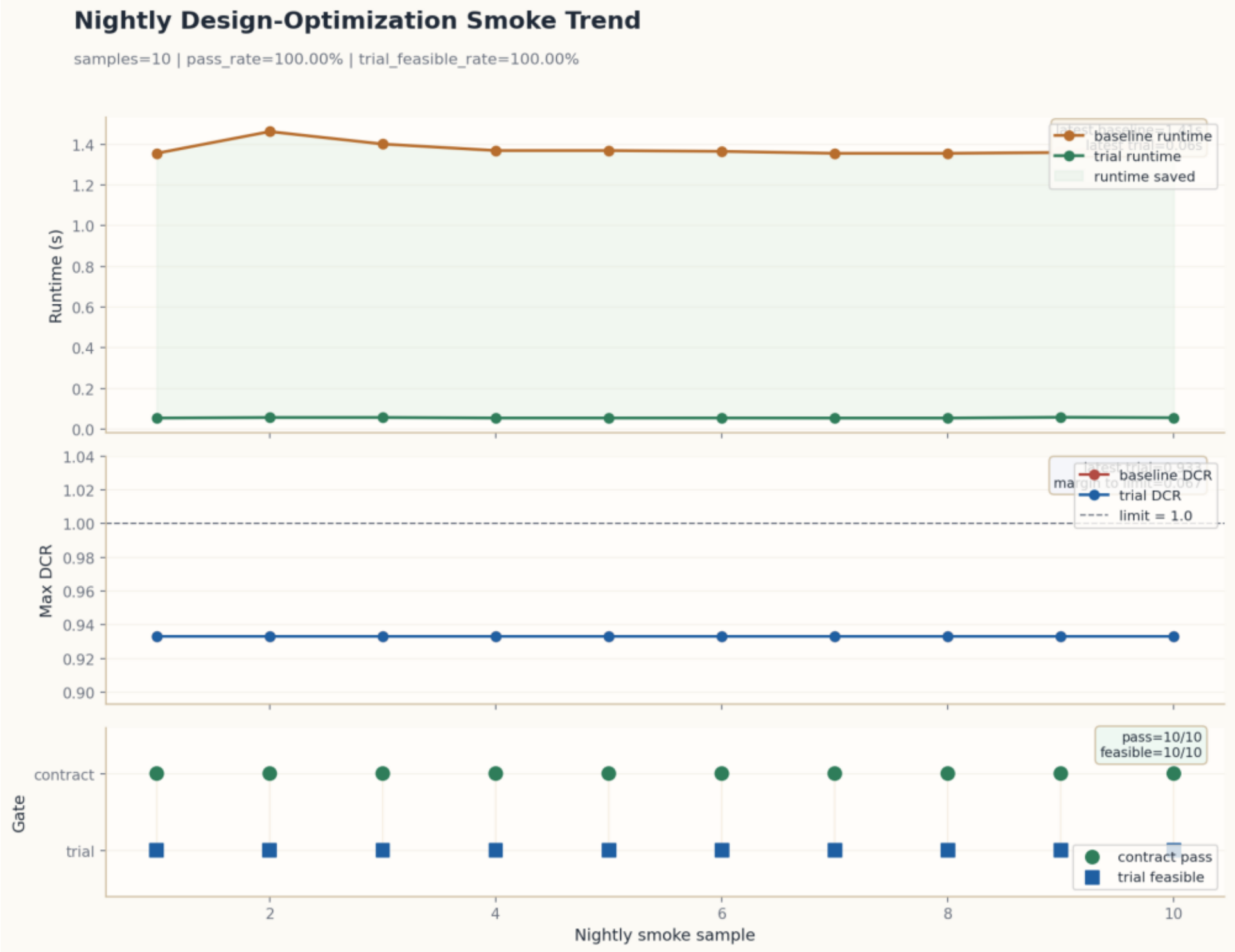
hazard=seismic | rows=48 | members=4 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001 | top_dcr=1.112

hazard=wind | rows=48 | members=4 | clauses=6 | combos=2 | top_clause=KDS-MOMENT-Y-001 | top_dcr=0.809

rule_family=strength | rows=120 | members=12 | hazards=3 | combos=2 | top_clause=KDS-MOMENT-Y-001 | top_dcr=1.216

rule_family=strength_interaction | rows=24 | members=12 | hazards=3 | combos=2 | top_clause=KDS-INT-FRAME-001 | top_dcr=1.137

Nightly Smoke Trend



Nightly Smoke Recent Samples

#6 2026-04-06T12:19:39.039411+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3646s | trial_runtime=0.0554s | trial_max_dcr=0.9332 | action=connect

#7 2026-04-06T12:23:56.108705+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3550s | trial_runtime=0.0550s | trial_max_dcr=0.9332 | action=connect

#8 2026-04-06T12:23:56.108705+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3550s | trial_runtime=0.0550s | trial_max_dcr=0.9332 | action=connect

#9 2026-04-06T12:32:45.009996+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.3590s | trial_runtime=0.0590s | trial_max_dcr=0.9332 | action=connect

#10 2026-04-06T13:54:47.180033+00:00 | pass=True | trial_feasible=True | baseline_runtime=1.4059s | trial_runtime=0.0567s | trial_max_dcr=0.9332 | action=connect

Measured Chain Category Trend

